



SUSTAINABLE COMMUNITY DEVELOPMENT

Sanitation, Water Supply & Land Use

Study notes on how safe waste management, reliable water systems, and planned land use work together to build healthy communities.

Sanitation

Water Supply

Land Use

01



Sanitation

Systems, practices, and facilities used to safely manage human waste, wastewater, and other forms of waste to protect health and the environment.

What is Sanitation?



Definition

Sanitation refers to the systems, practices, and facilities used to safely manage human waste, wastewater, and other forms of waste in order to protect human health and the environment.

It includes the safe collection, treatment, disposal, and management of human excreta, wastewater, solid waste, and drainage.

Why It Matters

- **Prevents diseases**

Reduces the spread of cholera, typhoid, dysentery, diarrhea, and other intestinal infections.

- **Protects water sources**

Stops sewage and waste from contaminating rivers, lakes, wells, and groundwater.

- **Improves public health**

Adequate toilets, drainage, and waste management mean healthier communities.

- **Supports dignity & the economy**

Safe toilets protect privacy, and good sanitation lowers healthcare costs.

Importance of Sanitation



Prevents Diseases

Reduces cholera, typhoid, dysentery, diarrhea, and intestinal infections by keeping waste away from water and food.



Protects Water Sources

Prevents sewage and human waste from contaminating rivers, lakes, wells, and groundwater.



Improves Public Health

Communities with adequate toilets, drainage, and waste management see better health outcomes.



Improves Environment

Prevents bad smells, pollution, blocked drains, and contamination of soil and water.



Dignity & Privacy

Clean, safe toilets protect privacy for women, children, older people, and people with disabilities.



Supports the Economy

Poor sanitation raises healthcare costs and lowers productivity through frequent illness.

Types of Sanitation

01



On-site Sanitation

Waste is treated or stored at or near the place where it is produced — e.g. pit latrines, septic tanks, and improved ventilated pit latrines.

02



Off-site Sanitation

Wastewater and sewage are transported through sewer systems to a treatment facility before disposal or reuse.

03



Solid Waste Sanitation

Collection, transportation, treatment, recycling, and safe disposal of household and commercial waste.

Problems & Improvements



Problems of Poor Sanitation

- Spread of waterborne diseases
- Contamination of drinking water
- Open defecation
- Bad smells and unpleasant surroundings
- Breeding of flies, mosquitoes, vectors
- Soil and water pollution
- Flooding from blocked drainage
- Reduced quality of life



Ways to Improve Sanitation

- Constructing and maintaining safe toilets
- Proper disposal of human waste
- Establishing sewage-treatment systems
- Improving garbage collection
- Providing proper drainage
- Promoting handwashing with soap
- Educating communities on hygiene
- Enforcing sanitation regulations

02



Water Supply

Obtaining, treating, storing, transporting, and distributing water for domestic, agricultural, industrial, and commercial use.

Sources of Water

Water supply is the process of obtaining, treating, storing, transporting, and distributing water for domestic, agricultural, industrial, and commercial purposes.



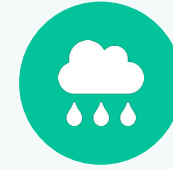
Surface Water

Rivers, lakes, streams, dams, and reservoirs. Easy to access but can be contaminated by sewage, agricultural chemicals, and industrial waste.



Groundwater

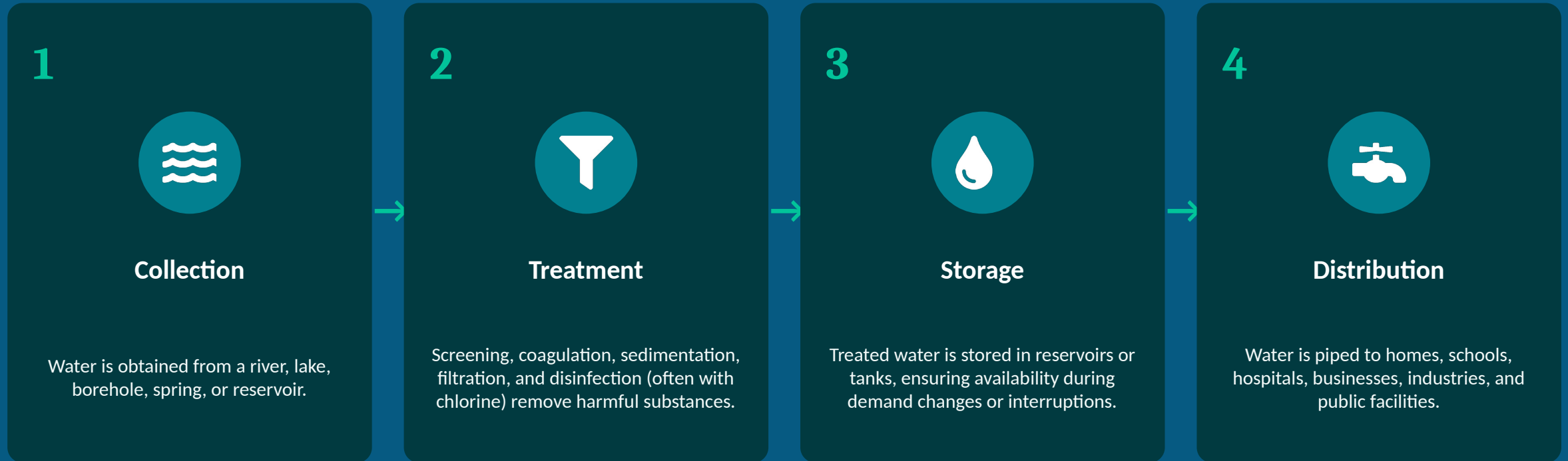
Boreholes, wells, and springs. Often protected from surface contamination, though it can still be polluted if waste enters the soil.



Rainwater

Collected from rooftops and stored in tanks. Especially useful where piped water systems are unavailable or unreliable.

Stages of Water Supply



Water Source → Collection → Treatment → Storage → Distribution → Consumer

Importance of Water Supply



Drinking

The human body requires water for normal functioning.



Cooking

Safe water is necessary for food preparation.



Personal Hygiene

Water is needed for bathing, handwashing, and cleaning.



Sanitation

Water supports toilet systems and wastewater management.



Agriculture

Essential for growing crops and raising animals.



Industry

Used for production, cooling, cleaning, and processing.



Economic Development

Reliable water supports businesses and employment.



Education

Schools need water for drinking, cleaning, toilets, and hygiene.

Problems & Improvements



Problems Affecting Water Supply

- Water pollution
- Drought and climate variability
- Population growth
- Rapid urbanization
- Poor infrastructure
- Leaking pipes
- High demand
- Poor management of water resources



Ways to Improve Water Supply

- Constructing more treatment facilities
- Expanding piped-water networks
- Protecting rivers, lakes, and groundwater
- Promoting rainwater harvesting
- Repairing leaking pipes
- Constructing protected boreholes/wells
- Improving water storage
- Educating on water conservation

03



Land Use

The way people use and manage land for different activities. Proper land-use planning ensures land is used efficiently without unnecessary environmental damage.

Major Types of Land Use (1/2)



Residential

Land used for housing: houses, apartments, hostels, and estates. Planning must consider roads, water, sanitation, electricity, and schools.



Commercial

Land for buying and selling goods and services: shops, markets, shopping centers, banks, restaurants, and offices.



Industrial

Used for manufacturing, processing, and storage. Needs strong infrastructure and regulation to control pollution.



Agricultural

Used for crop production, livestock, fishing, and forestry. Provides food, employment, raw materials, and income.

Major Types of Land Use (2/2)



Recreational

Land for leisure, sports, and entertainment: parks, playgrounds, sports fields, and gardens.



Institutional

Land for public and private institutions: schools, universities, hospitals, government offices, and police stations.



Transportation

Land needed for movement: roads, railways, airports, bus terminals, and parking areas.



Conservation

Protects natural ecosystems: forest reserves, wetlands, national parks, wildlife areas, and water catchments.

How the Three Connect

These three areas are closely connected. Poor planning in one affects the others.



If land is poorly planned, houses may be built too close to wetlands, rivers, or water sources — causing pollution and flooding. If a community has good water supply but poor sanitation, sewage can contaminate the very water sources people depend on.

Good land-use planning must incorporate adequate water supply and sanitation systems.

A Growing Community Near a River

Imagine a community that builds without proper planning:

1

People obtain water directly from the river.

2

Toilets are constructed too close to the river.

3

Waste is dumped near the river.

4

During rainfall, waste is washed into the river.

5

The river becomes contaminated.

6

People using the water develop diseases.



CONCLUSION

Fundamentals of Sustainable Community Development

Sanitation, water supply, and land use are fundamental components of sustainable community development. Sanitation protects people from diseases and environmental pollution, while water supply provides safe and reliable water for domestic, agricultural, industrial, and commercial activities. Land use determines how available land is allocated for housing, agriculture, industry, transportation, recreation, institutions, and conservation.

Proper planning and management of all three are necessary to create healthy, safe, productive, and sustainable communities.